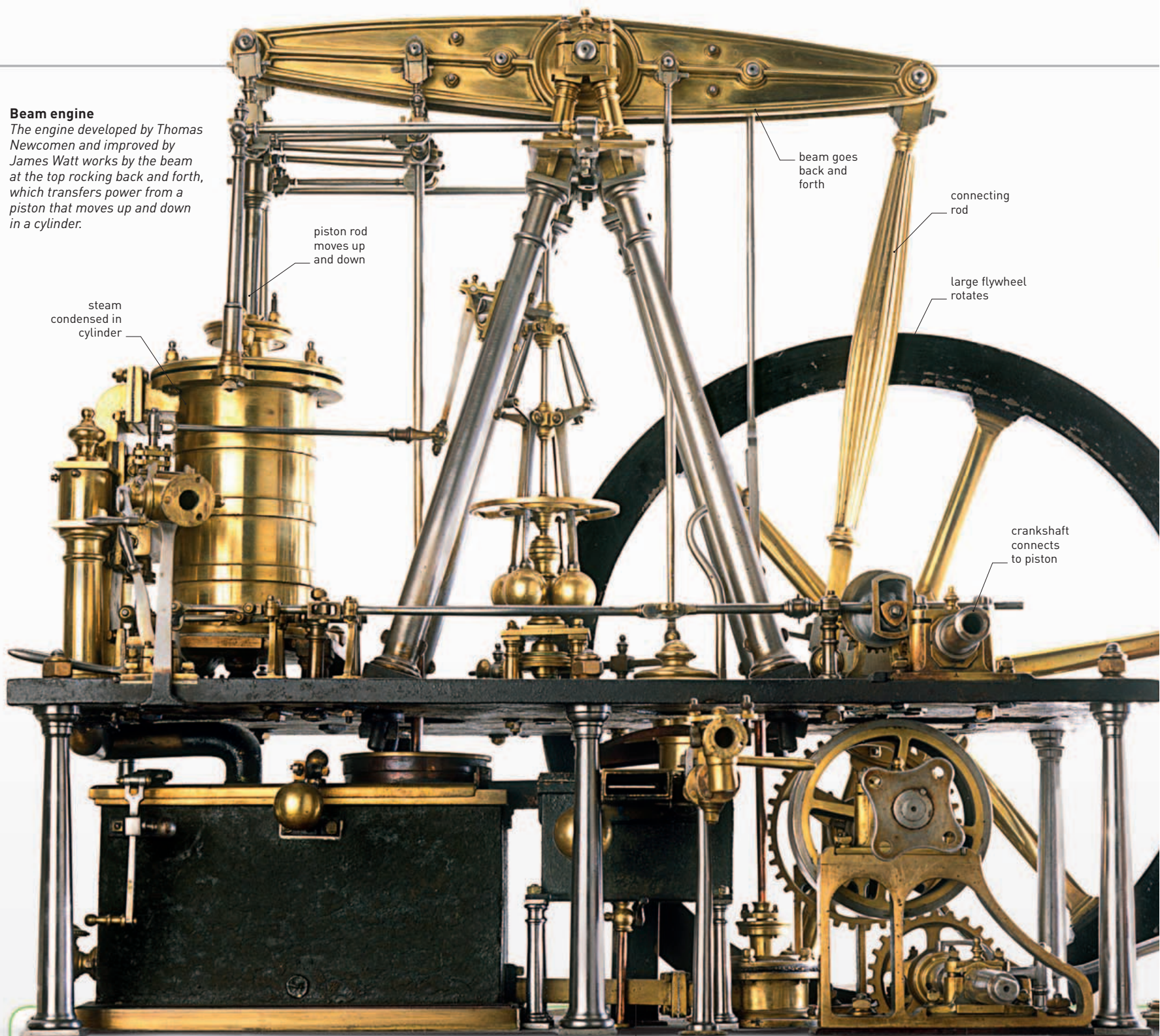


Beam engine

The engine developed by Thomas Newcomen and improved by James Watt works by the beam at the top rocking back and forth, which transfers power from a piston that moves up and down in a cylinder.



1679

The first boiler

French inventor Denis Papin designs a device that can convert liquid to vapor, making it the first pressure cooker.

Papin's steam digester



1698

The high-pressure steam engine

In England, Thomas Savery uses steam power to create "The Miner's Friend" to pump water out of coal mines, although it was not a success.



Newcomen's atmospheric engine

1765

James Watt's engine

Scottish inventor James Watt makes improvements to the Newcomen engine by adding a condenser, and develops an engine that rotates a shaft instead of pumping.

1801-04

Trevithick's engine

English mine engineer Richard Trevithick develops a smaller, lighter steam engine and puts it on wheels, creating a "road locomotive."

1st century CE

Hero's engine

The Greek scientist Hero describes an aeolipile, which has a rotating ball that is spun by jets of steam.



The Hero engine

1712

Newcomen's engine

Thomas Savery joins forces with Thomas Newcomen and they create the much-improved atmospheric steam pumping engine.

1769-70

The steam car

In France, Nicholas Cugot invents a road vehicle that can run on steam by converting it into piston action and rotary motion.

1802-07

The steamboat

In the US, Robert Fulton applies steam power to a passenger boat, and it proves a success in sailing against currents.